x0xb0x Mainboard Beta Release (3/8/2005)

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CPU

File: p02_cpu.kicad_sch

LEDs

File: p05_leds.kicad_sch

VCA

File: p08_vca.kicad_sch

Rotary encoders

File: p03_encoders.kicad_sch

CV and VCO

File: p06_cv_vco.kicad_sch

Envelope

File: p09_env.kicad_sch

Tact switches

File: p04_switches.kicad_sch

VCF

File: p07_vcf.kicad_sch

Mixer and headphone

File: p10_mixer.kicad_sch

x0xb0x mainboard

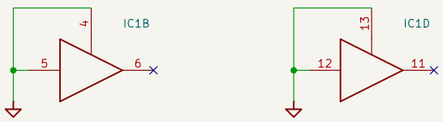
Sheet: /

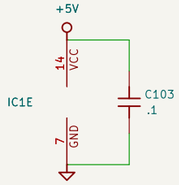
File: mainboard.kicad_sch

Title: x0xb0x mainboard

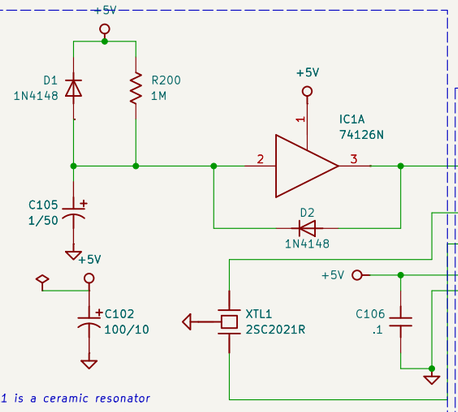
Size: A3	Date:	**Rev: K30**
KiCad E.D.A. 10.0.7		Id: 1/10

Unused gates of IC1



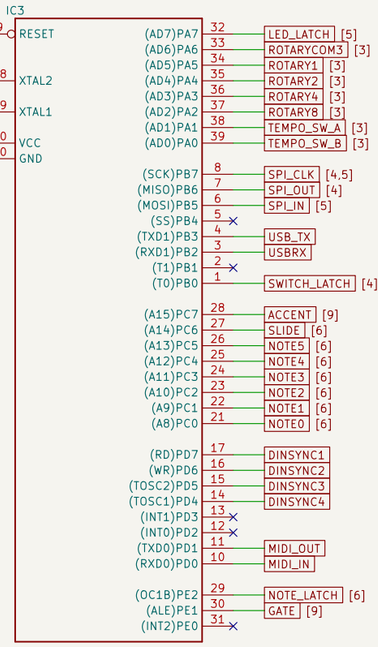


Reset and clock

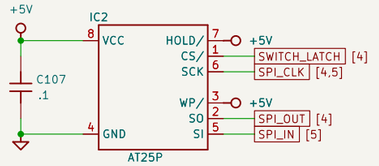


XTL1 is a ceramic resonator

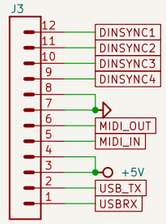
MCU



EEPROM



To the I/O board



Mounting holes



3.3 mm holes: on the board only, not on the original sheet

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Drawn from the original Eagle schematic; nets checked against the board copper.

x0xb0x mainboard

Sheet: /CPU/
File: p02_cpu.kicad_sch

Title: CPU

Size: A3	Date:	**Rev: K30**
KiCad E.D.A. 10.0.7		Id: 2/10

Rotary Encoders

function select

S4

D6 1N4148
D5 1N4148
D4 1N4148
D3 1N4148

P$8
P$4
P$2
P$1
C

MT1 M1
MT2 M2
MT3 M3
MT4 M4

GH3074

R255 10K

Q5 2SC536F

R254 10K

bank select

S3

D10 1N4148
D9 1N4148
D8 1N4148
D7 1N4148

P$8
P$4
P$2
P$1
C

MT1 M1
MT2 M2
MT3 M3
MT4 M4

GH3074

[2] ROTARYCOM3

[2] ROTARY8
[2] ROTARY4
[2] ROTARY2
[2] ROTARY1

tempo

J1
TEMPO_SW [4]

S2
SW1
SW2
P$2 M2
P$1 M1

J2
+5V

R233 10K

R270 10K
R269 10K

A
B
C

PEC11-S

R235 220k
R236 220k

TEMPO_SW_B [2]
TEMPO_SW_A [2]

C100 .001
C101 .001

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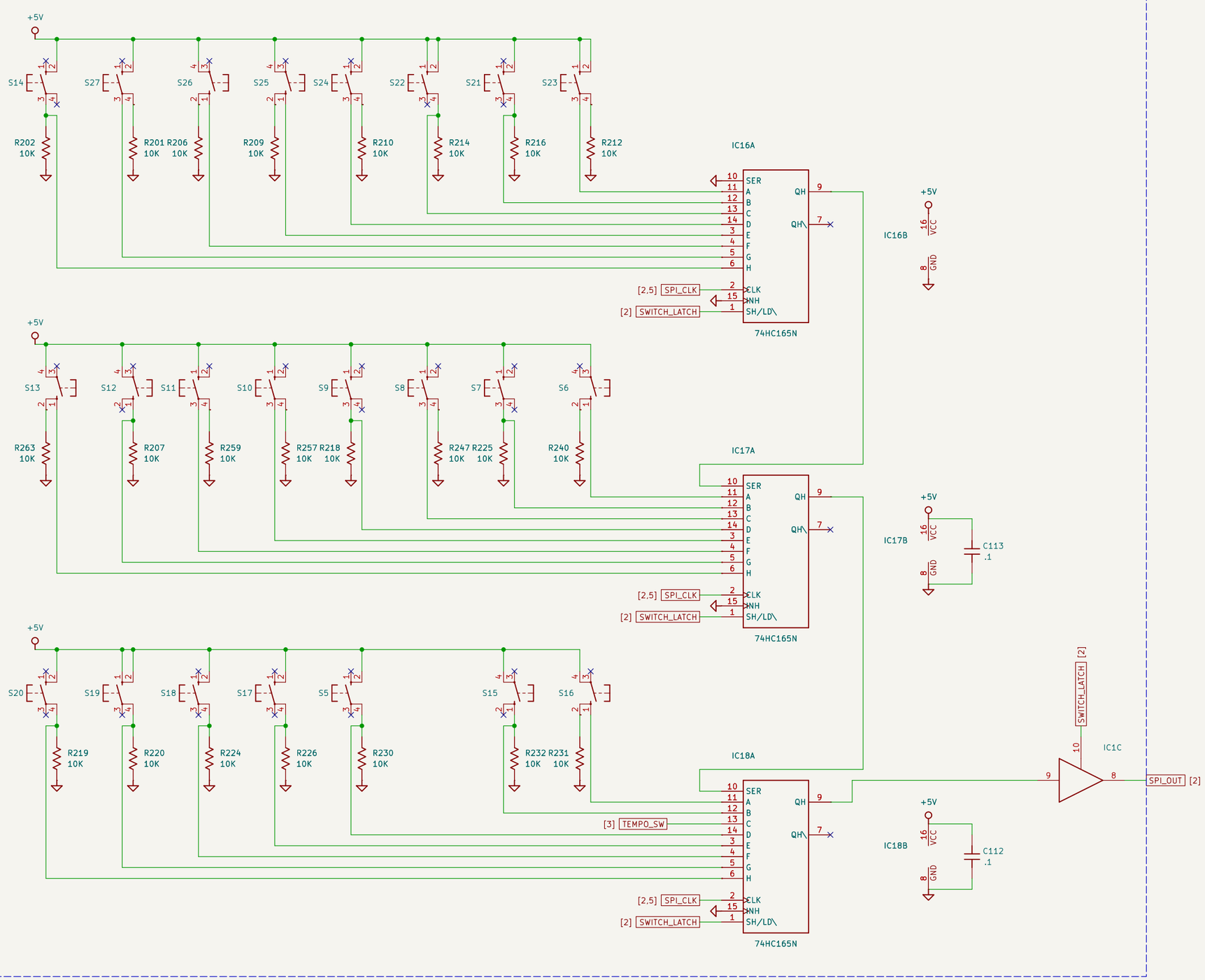
x0xb0x mainboard

Sheet: /Rotary encoders/
File: p03_encoders.kicad_sch

Title: Rotary encoders

Size: A3	Date:	**Rev: K30**
KiCad E.D.A. 10.0.7		Id: 3/10

Tact Switches



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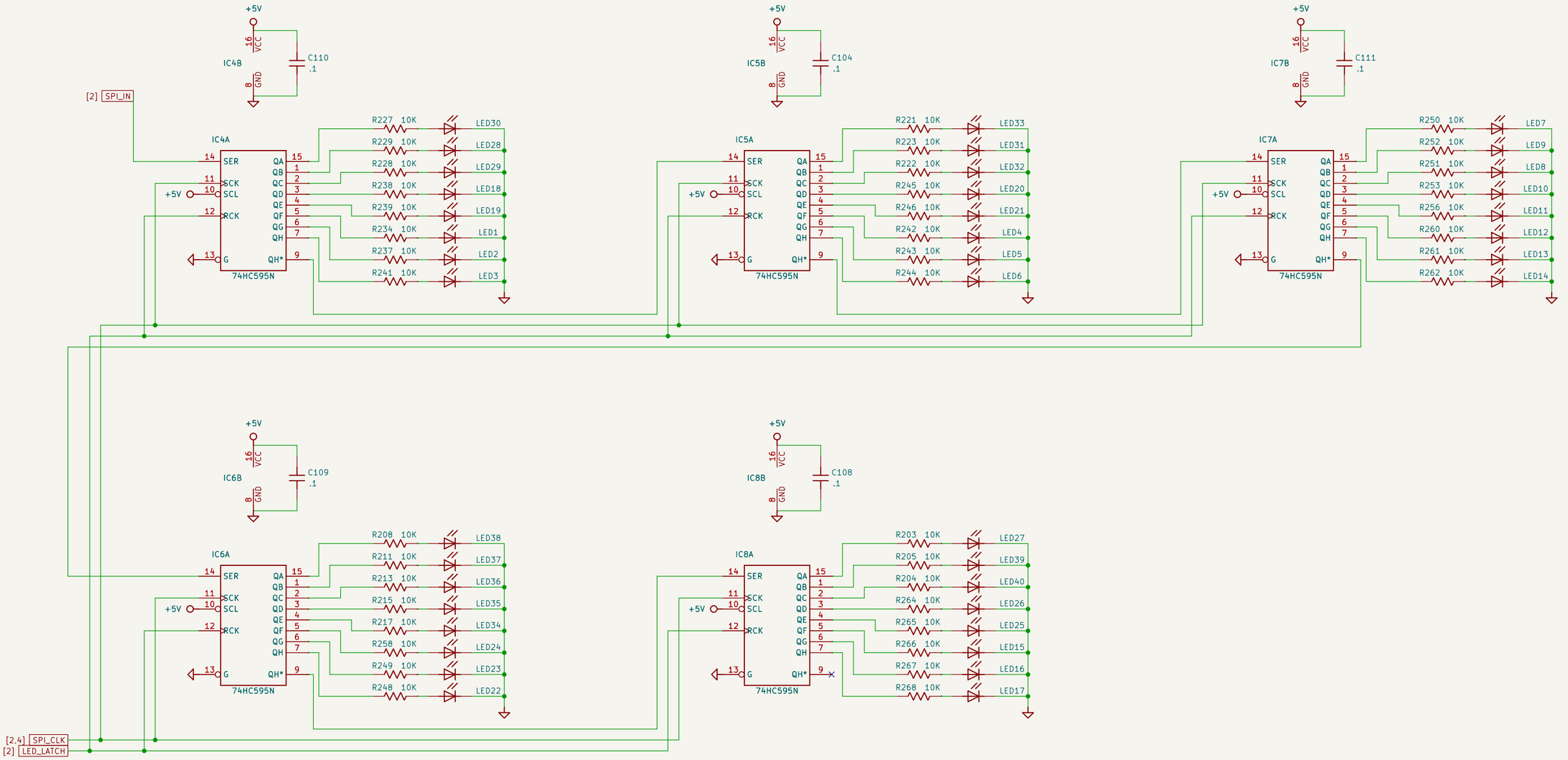
x0xb0x mainboard

Sheet: /Tact switches/
File: p04_switches.kicad_sch

Title: Tact switches

Size: A3	Date:	**Rev: K30**
KiCad E.D.A. 10.0.7		Id: 4/10

LEDs



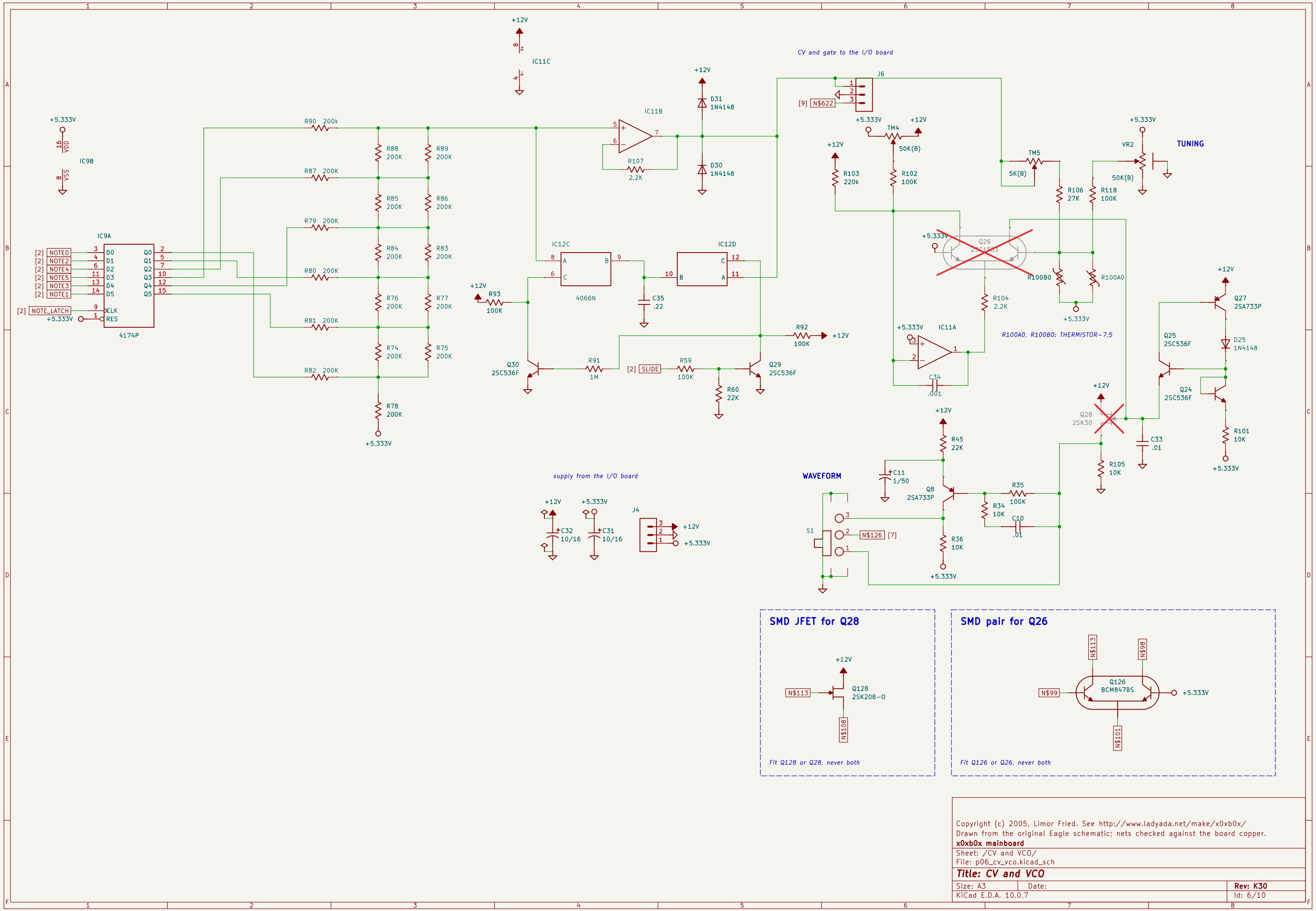
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x0xb0x mainboard

Sheet: /LEDs/
File: p05_leds.kicad_sch

Title: LEDs

Size: A3	Date:	**Rev: K30**
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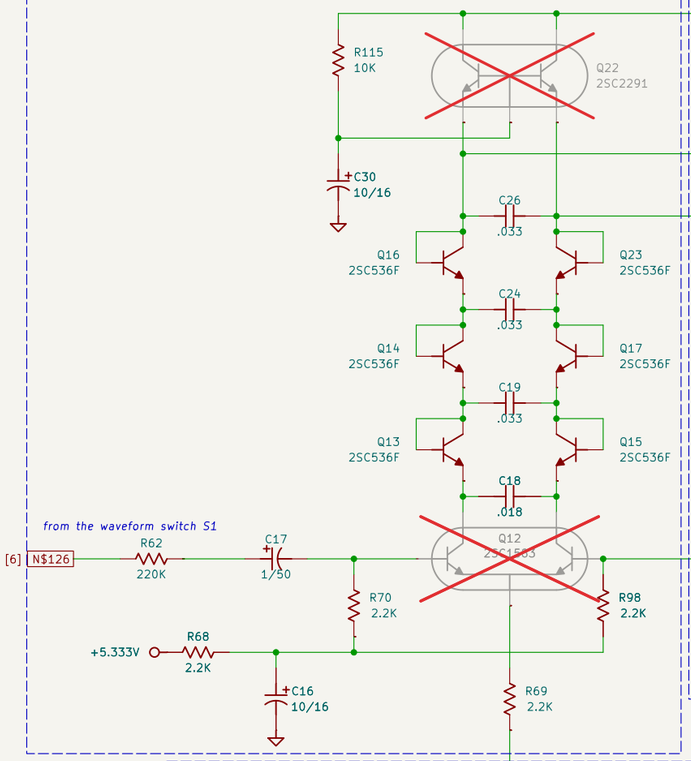
x0xb0x mainboard

Sheet: /CV and VCO/
File: p06_cv_vco.kicad_sch

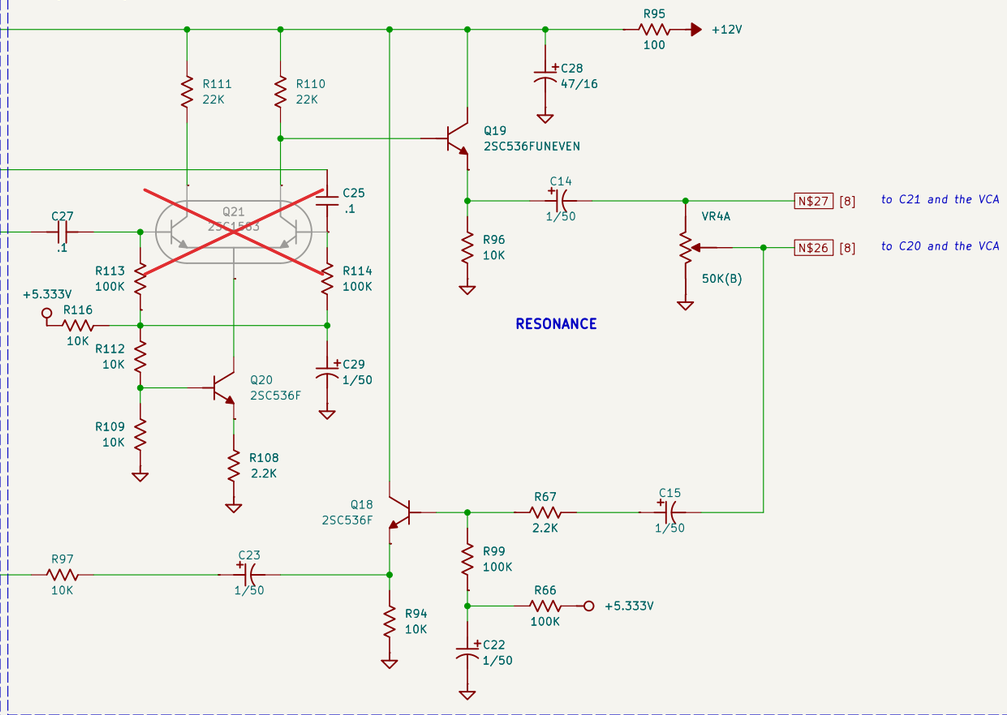
Title: CV and VCO

Size: A3	Date:	**Rev: K30**
KiCad E.D.A. 10.0.7		Id: 6/10

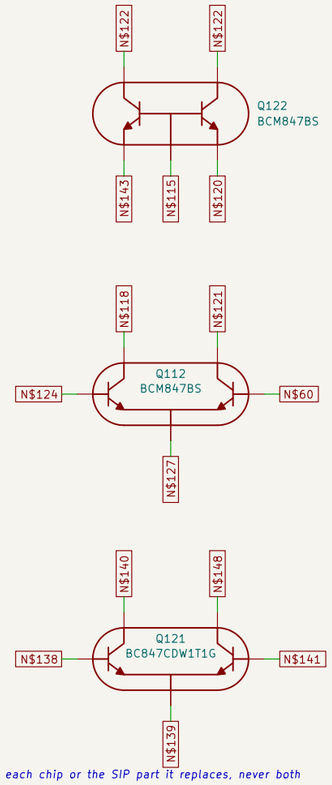
Diode ladder



Output amplifier and resonance

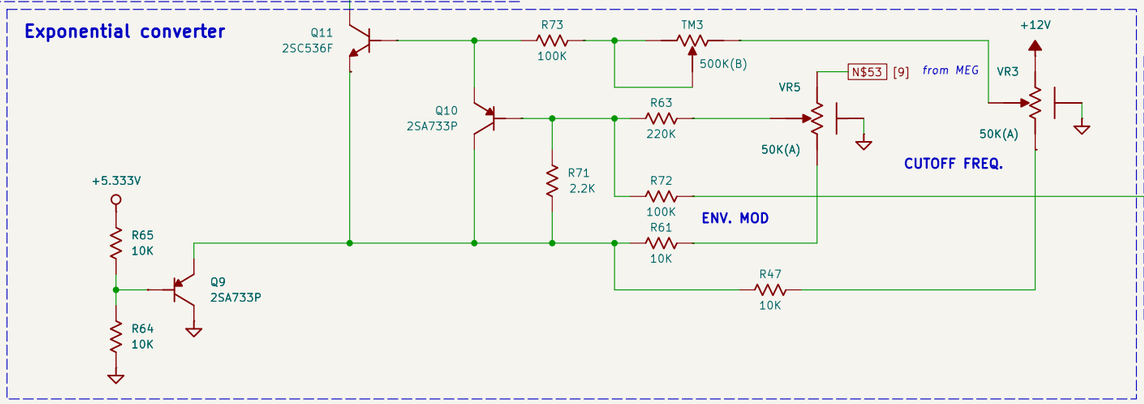


SMD pairs

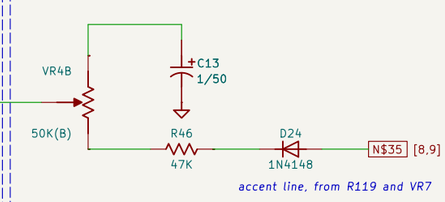


Fit each chip or the SIP part it replaces, never both

Exponential converter



Accent sweep



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x0xb0x mainboard

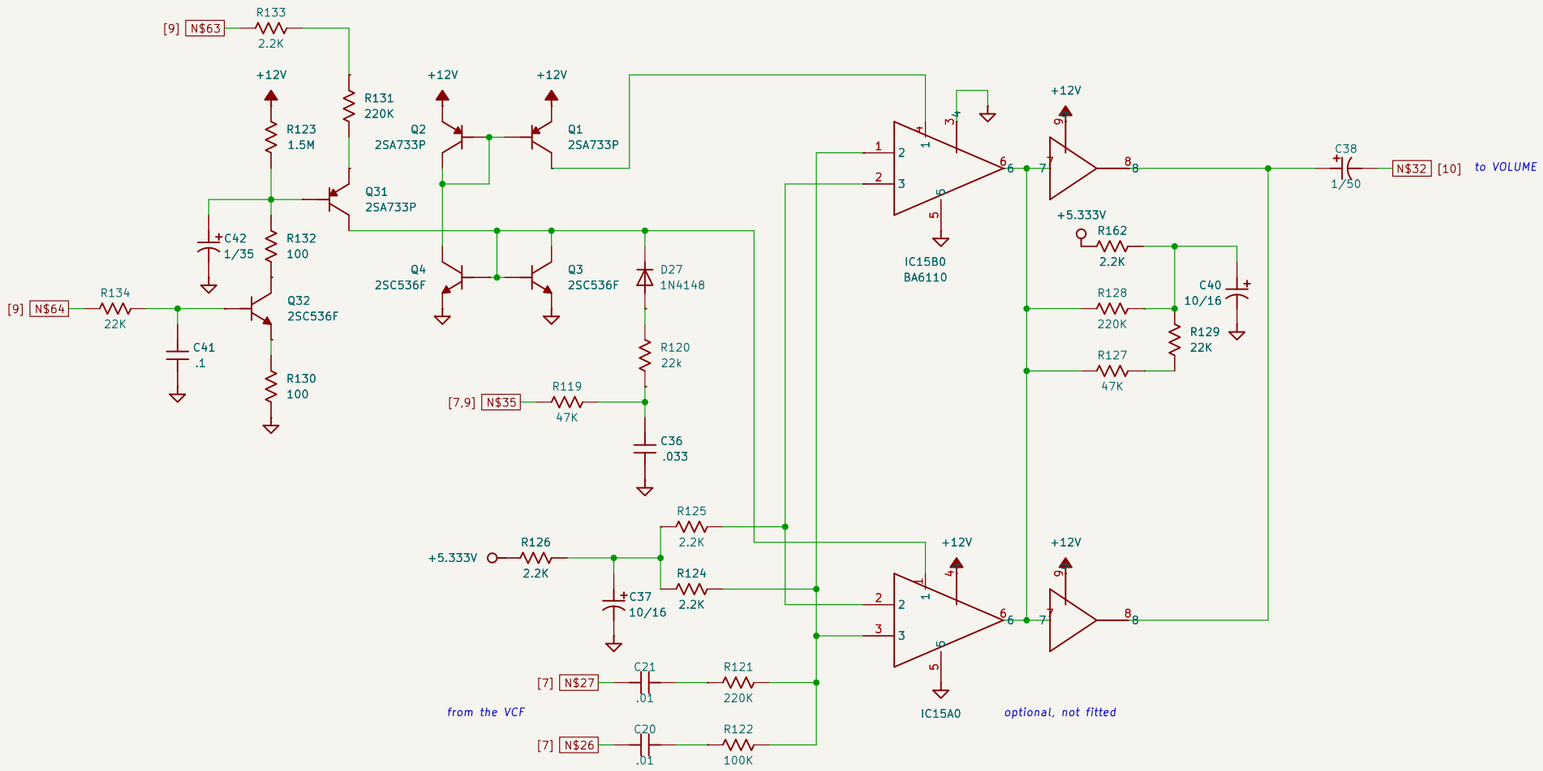
Sheet: /VCF/
File: p07_vcf.kicad_sch

Title: VCF

Size: A3	Date:	**Rev: K30**
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VCA

the envelope, turned into a current



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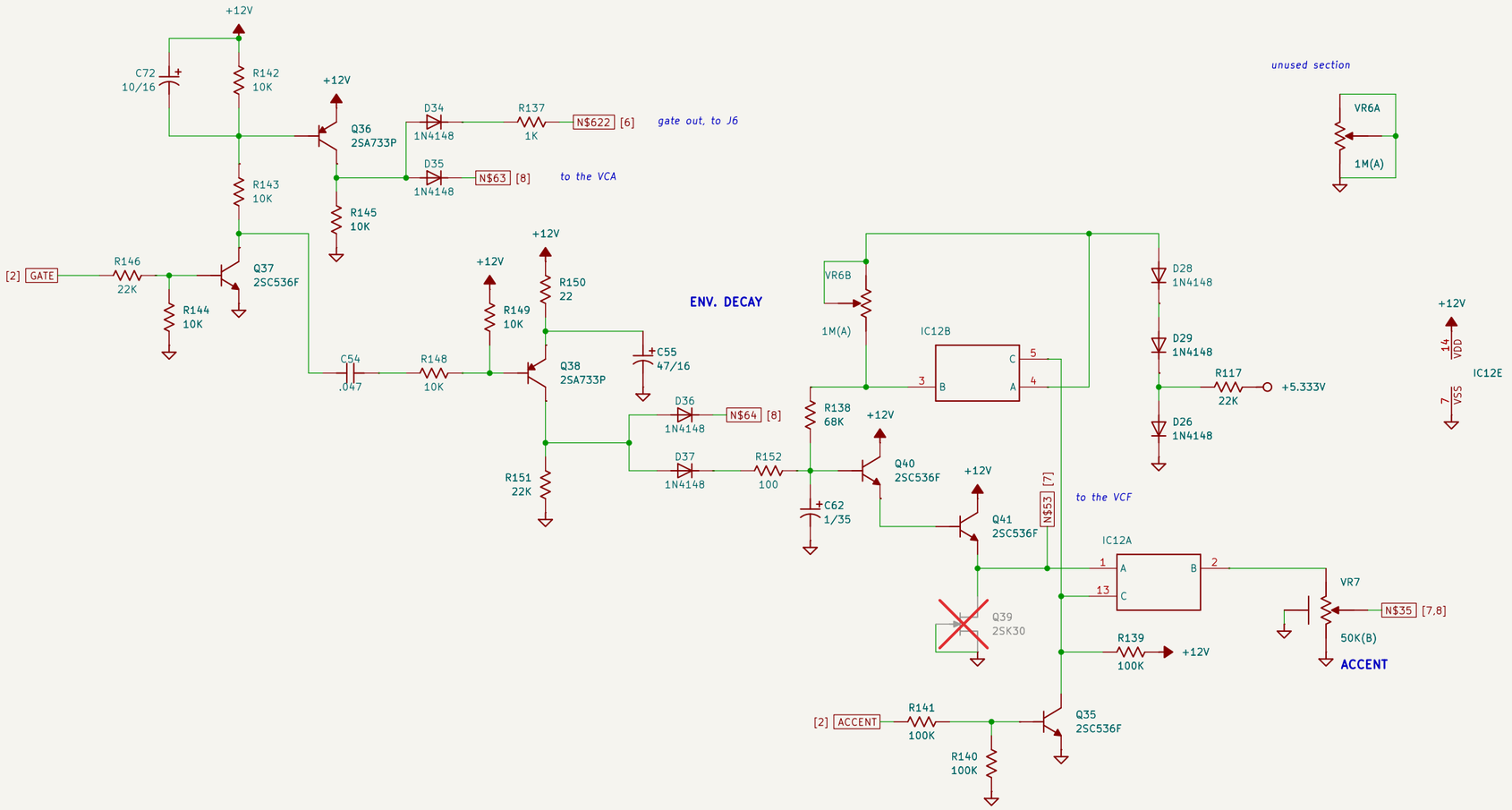
x0xb0x mainboard

Sheet: /VCA/
File: p08_vca.kicad_sch

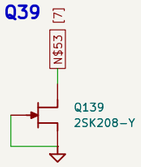
Title: VCA

Size: A3	Date:	**Rev: K30**
KiCad E.D.A. 10.0.7		Id: 8/10

ENV



SMD JFET for Q39



Fit Q139 or Q39, never both

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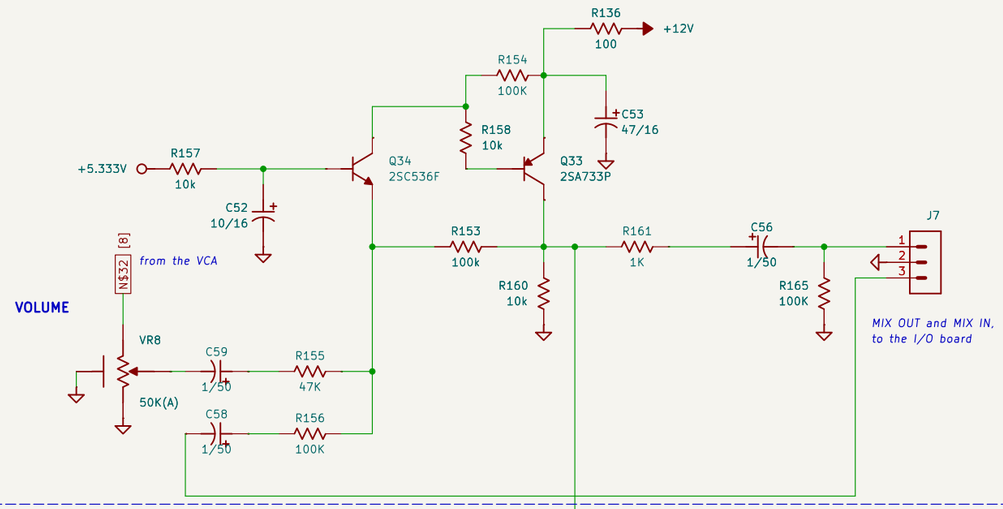
x0xb0x mainboard

Sheet: /Envelope/
File: p09_env.kicad_sch

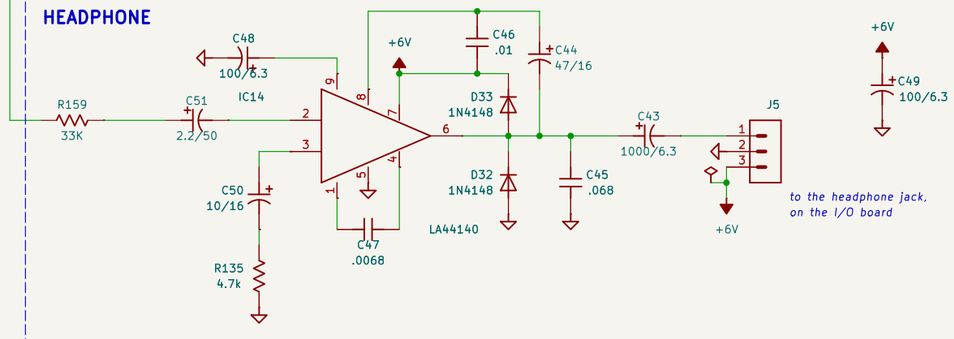
Title: Envelope

Size: A3	Date:	**Rev: K30**
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MIXER



HEADPHONE



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x0xb0x mainboard

Sheet: /Mixer and headphone/
File: p10_mixer.kicad_sch

Title: Mixer and headphone

Size: A3	Date:	**Rev: K30**
KiCad E.D.A. 10.0.7		Id: 10/10